

LAB 8

Server Configuration – DHCP, DNS and Web Server in Cisco Packet Tracer

OBJECTIVES

- To configure a server in Cisco Packet Tracer to provide DHCP, DNS, and Web (HTTP) services.
- To understand the process by which DHCP assigns IP addresses to client devices.
- To learn how DNS resolves domain names into corresponding IP addresses within a network.
- To verify the functionality of these services by accessing the web server from client devices.

THEORY

1. Dynamic Host Configuration Protocol (DHCP)

DHCP is a network protocol that automatically assigns IP addresses, subnet masks, default gateways, and DNS information to client devices. Instead of manually configuring each device, DHCP ensures efficient and error-free IP address management. In Cisco Packet Tracer, enabling DHCP on a server allows PCs and other hosts to request and obtain IP configurations dynamically. This reduces administrative overhead and prevents conflicts caused by duplicate IP addresses.

2. Domain Name System (DNS)

DNS is responsible for translating human-readable domain names (e.g., `www.example.com`) into machine-readable IP addresses. Without DNS, users would need to remember numerical IP addresses to access resources. In Packet Tracer, configuring DNS on a server allows client devices to resolve domain names to the correct IP address of the web server. This simulates real-world internet behavior, where DNS servers act as the “phonebook” of the network.

3. Web Server (HTTP Service) :

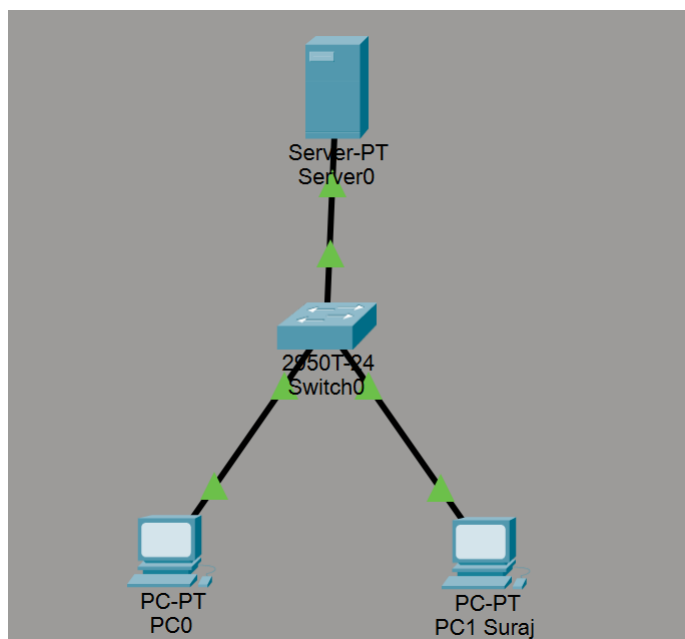
A web server hosts websites and responds to client requests using the Hypertext Transfer Protocol (HTTP). When a client enters a domain name in its browser, the DNS resolves it to the server’s IP, and the HTTP service delivers the requested web page. In Packet Tracer, enabling the HTTP service on the server allows PCs to access and display web content, verifying that DHCP and DNS are functioning correctly.

4. Integration of Services

- DHCP, DNS, and HTTP services work together to provide seamless connectivity:
 - DHCP assigns IP addresses to clients.
 - DNS resolves domain names to the server's IP.
 - The HTTP server delivers web content to the client.
- Testing these services in Packet Tracer demonstrates how real networks provide automated configuration, name resolution, and web access.

NETWORK TOPOLOGY

This network topology illustrates a basic star configuration where all devices connect through a central switch. The server provides DHCP, DNS, and Web services, while the two PCs act as clients to test connectivity and service functionality. This setup demonstrates how centralized resources can efficiently manage IP addressing, domain resolution, and web hosting in a small network.



CONFIGURATION

IP Addressing Scheme

Device	IP Address	Subnet Mask	Gateway
Server	192.168.1.2	255.255.255.0	192.168.1.1
PC	DHCP	DHCP	DHCP

For PCs,

Device	IPv4 Address	Subnet Mask	Default Gateway	DNS Server
PC0	192.168.1.1	255.255.255.0	0.0.0.0	192.168.1.2
PC1	192.168.1.3	255.255.255.0	0.0.0.0	192.168.1.2

Note: The PCs IP address are dynamically generated via DHCP server. The above mentioned is obtained from DHCP server (no manual registering).

SERVER CONFIGURATION

Step 1 : Assign Static IP to Server

1. Click on Server
2. Go to Desktop → IP Configuration
3. Configure the following:
 - IP Address: 192.168.1.2
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.1.1
 - DNS Server: 192.168.1.2

DHCP Configuration :

Step 2 : Enable DHCP Service

1. Go to Server → Services → DHCP
2. Turn DHCP: ON

Step 3 : Create DHCP Pool

Field	Value
Pool Name	LAB_Pool
Default Gateway	192.168.1.1
DNS Server	192.168.1.2

Start IP Address	192.168.1.10
Subnet Mask	255.255.255.0
Maximum Users	50

Click Add

Verification :

On each PC:

- Go to Desktop → IP Configuration
- Select DHCP

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1
```

```
Pinging 192.168.1.1 with 32 bytes of data:
```

```
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Fig: Ping Test from PC1 TO PC0

DNS Server Configuration :

Step 4 : Enable DNS Service

1. Go to Server → Services → DNS
2. Turn DNS: ON

Step 5 : Add DNS Records

- Name: www.hcoe.com
- Type: A record
- Address: 192.168.1.2

Click Add

Verification :

On each PC:

- Open Command Prompt
- Ping: www.hcoe.com

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.hcoe.com
```

```
Pinging 192.168.1.2 with 32 bytes of data:
```

```
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=4ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms
```

Fig: Ping Test from PC0

```
C:\>ping www.hcoe.com
```

```
Pinging 192.168.1.2 with 32 bytes of data:
```

```
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Fig: Ping Test from PC1 Suraj

Web Server Configuration :

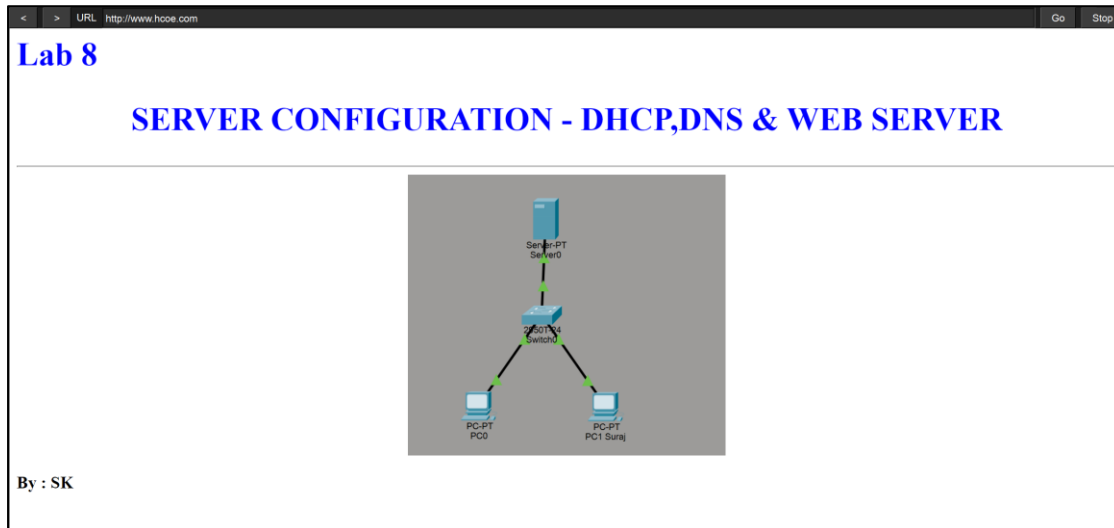
Step 6 : Enable HTTP Service

1. Go to Server → Services → HTTP
2. Turn HTTP: ON
3. Edit index.html (optional)

Verification :

On PC:

- Go to Desktop → Browser
- Type: <http://www.hcoe.com>



DISCUSSION

In this lab, a server was configured to provide DHCP, DNS, and Web services within a star topology network in Cisco Packet Tracer. The DHCP service successfully automated IP address assignment to client PCs, eliminating manual configuration. DNS resolved domain names to the server's IP, enabling clients to access the hosted web page through a browser. The integration of these services demonstrated how centralized management simplifies network operations and ensures seamless communication between devices.

CONCLUSION

The lab confirmed that DHCP, DNS, and Web services can be effectively implemented and tested in Cisco Packet Tracer. DHCP streamlined IP allocation, DNS provided name resolution, and the web server delivered content to clients. Together, these services validated the importance of centralized server configuration in maintaining efficient and user-friendly network environments.